Supplementary Material S1
Search Strings by Platform

Facial Emotion Recognition through Machine Learning in Children with Autism Spectrum Disorder:
A systematic scoping review of open-access computational evidence and of the gaps toward therapeutic monitoring

1. Summary of execution

The three main searches were executed on 15 February 2026. The three strings share in full the same Boolean structure and the same descriptors, organised into three conceptual blocks joined by AND: population (ASD), concept (facial emotional expressiveness) and method (machine learning). The only difference across platforms is one of field syntax and temporal restriction, given that Web of Science and IEEE Xplore do not support the PUBYEAR operator of Scopus and employ their own fields, as documented in Section 2.3 of the manuscript.

Platform	Temporal range	Records retrieved	Execution date
Scopus	2019–2025	935	15/02/2026
Web of Science Core	2018–2026	942	15/02/2026
IEEE Xplore	2018–2026	38	15/02/2026
Total		**1,915**	

Distribution by platform at each stage of the selection process:

Stage	Scopus	WoS	IEEE	Total
Identification (records retrieved)	935	942	38	1,915
Screening (after duplicate removal)	642	258	13	913
Full-text retrieval	60	17	8	85
Meeting formal criteria	25	7	2	34

Of the 34 studies meeting the formal inclusion criteria, 3 were excluded following the post-search editorial status verification described in Section 2.7 of the manuscript. Of the remaining 31, 24 constitute the eligible corpus and 7 are documented as contextual evidence outside the eligible set, as detailed in Section 2.4 of the manuscript. A complementary search in ACM Digital Library retrieved no additional eligible studies; all five ACM-native records failed CI(4) (open access). Forward and backward citation chasing identified five candidate studies that preliminarily meet the eligibility criteria; their full-text screening is pending for the submission version. No studies were incorporated through expert consultation.

The 25 studies retrieved in Scopus correspond entirely to the 2019–2025 period; Web of Science contributed the two studies published in 2018 and IEEE Xplore the single one published in 2026, a difference explained by the platform-specific temporal restriction documented in Section 2.3 of the manuscript. The three strings covered the year 2020 within their temporal range and retrieved records from that year, but none passed the screening and eligibility phases, so 2020 appears with a value of zero in the annual distribution of the corpus; that column is retained in the corresponding table and figure of the manuscript to preserve the continuity of the temporal series. The disaggregation by platform reported in this table corresponds to the internal extraction record; the PRISMA diagram of the manuscript reports it only in the identification stage, since after deduplication the removal of a duplicate record is not attributable to any individual platform.

2. Conceptual blocks common to the three platforms

Block	Descriptors joined by OR
Population	autism; ASD; autism spectrum; autistic
Concept	emotion; facial expression; affective; social communication
Method	machine learning; deep learning; artificial intelligence; computer vision; neural network; classification; recognition

The three blocks are combined with AND. The filters applied across the three platforms are: document type restricted to journal article and conference paper; open access; and English language.

3. String executed in Scopus

Temporal range: 2019–2025. Records retrieved: 935.

```
TITLE-ABS-KEY(("autism" OR "ASD" OR "autism spectrum" OR "autistic") AND ("emotion" OR "facial
expression" OR "affective" OR "social communication") AND ("machine learning" OR "deep learning"
OR "artificial intelligence" OR "computer vision" OR "neural network" OR "classification"
OR "recognition")) AND PUBYEAR > 2018 AND PUBYEAR < 2026 AND (LIMIT-TO(DOCTYPE,"ar") OR
LIMIT-TO(DOCTYPE,"cp")) AND (LIMIT-TO(OA,"all")) AND (LIMIT-TO(LANGUAGE,"English"))
```

The TITLE-ABS-KEY field searches simultaneously in title, abstract and keywords. The filter PUBYEAR $>$ 2018 AND PUBYEAR $<$ 2026 restricts retrieval to the 2019–2025 period inclusive.

4. String executed in Web of Science Core Collection

Temporal range: 2018–2026. Records retrieved: 942.

```
TS=(("autism" OR "ASD" OR "autism spectrum" OR "autistic") AND ("emotion" OR "facial expression"
OR "affective" OR "social communication") AND ("machine learning" OR "deep learning" OR
"artificial intelligence" OR "computer vision" OR "neural network" OR "classification" OR
"recognition")) AND PY=(2018-2026) AND DT=(Article OR Proceedings Paper) AND LA=(English)
```

The TS= (Topic) field is the functional equivalent of TITLE-ABS-KEY in Scopus: it searches title, abstract, author keywords and Keywords Plus. PY= restricts by publication year, DT= by document type and LA= by language. The open-access filter was applied as an interface refinement (Open Access: All), given that Web of Science does not support it as an operator within the string.

5. String executed in IEEE Xplore Digital Library

Temporal range: 2018–2026. Records retrieved: 38.

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(("All Metadata":"autism" OR "All Metadata":"ASD" OR "All Metadata":"autism spectrum" OR
"All Metadata":"autistic") AND ("All Metadata":"emotion" OR "All Metadata":"facial expression"
OR "All Metadata":"affective" OR "All Metadata":"social communication") AND ("All Metadata":"machine
learning" OR "All Metadata":"deep learning" OR "All Metadata":"artificial intelligence"
OR "All Metadata":"computer vision" OR "All Metadata":"neural network" OR "All Metadata":"classifica
OR "All Metadata":"recognition"))
```

Executed through Command Search. IEEE Xplore does not support temporal restriction, document type, open access or language operators within the string, so the four filters were applied as interface refinements: publication date range 2018–2026; content type restricted to Journals and Conferences; Open Access; and English language. The All Metadata field is broader than TITLE-ABS-KEY in Scopus, as it additionally includes IEEE indexing terms.

6. Pilot searches not incorporated into the corpus

During the iterative development of the strategy, pilot searches were executed with therapeutic intervention and gamification terminology, in order to verify whether the main string adequately captured the therapeutic monitoring component:

- serious game AND autism AND emotion recognition
- therapeutic monitoring AND ASD AND machine learning
- mHealth AND autism AND facial expression

These searches jointly retrieved fewer than 2% of additional records not overlapping with the main string. This result motivated the decision to prioritise the core terms of the domain, and indicates that the therapeutic monitoring literature published in the three selected databases rarely employs such terminology as a primary descriptor in titles and abstracts. The limitation deriving from this decision is documented in Section 2.8 of the manuscript.

7. Complementary procedures

The following complementary procedures were executed after the main search, as documented in Section 2.3 of the manuscript:

- Search in ACM Digital Library with equivalent Boolean structure and descriptors. Five ACM-native records were identified; none was available under open access and all were excluded per CI(4).
- Backward citation tracking on the reference lists of the included studies and forward citation tracking on the works citing them, conducted through platform citation functions and two recent systematic reviews. Five candidate studies were identified that preliminarily meet the eligibility criteria; their full-text screening and, if applicable, incorporation into the corpus will be documented in the submission version.

The following procedures were not executed:

- Formal PRESS (Peer Review of Electronic Search Strategies) review by a specialist librarian external to the team.
- Searching for similar articles through the recommendation functions of the platforms.
- Updating of the search after 15 February 2026.
- Systematic exploration of grey literature (preprints, technical reports and dissertations).